

CS23331-DAA-2025-CSE / 1-DP-Playing with Numbers



1-DP-Playing with Numbers

Started on	Wednesday, 7 October 2026, 11:37 PM
State	Finished
Completed on	Wednesday, 7 October 2026, 11:46 PM
Time taken	8 mins 47 secs

Time taken	8 mins 47 secs
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 10.00 out of 10.00 | [Flag question](#)

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3. Write any efficient algorithm to find the possible ways.

Example 1:

Input: 6

Output: 6

Explanation: There are 6 ways to 6 represent number with 1 and 3

1+1+1+1+1+1

3+3

1+1+1+3

1+1+3+1

1+3+1+1

3+1+1+1

1+1+3+1

1+3+1+1

3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      long long dp[n+1];
6      dp[0]=1;
7      dp[1]=1;
8      dp[2]=1;
9      for(int i=3;i<=n;i++){
10         dp[i]=dp[i-1]+dp[i-3];
11     }
12     printf("%lld",dp[n]);
13 }
```

	Input	Expected	Got	
✓	6	6	6	✓
✓	25	8641	8641	✓
✓	100	24382819596721629	24382819596721629	✓

Passed all tests! ✓

Correct

Marks for this submission: 10.00/10.00.



2-Finding Duplicates- $O(n)$ Time Complexity, $O(1)$ Space Complexity

Started on	Friday, 9 October 2026, 12:16 AM
State	Finished
Completed on	Friday, 9 October 2026, 12:48 AM
Time taken	31 mins 50 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00 | [Flag question](#)

Find Duplicate in Array.

Given a read only array of n integers between 1 and n , find one number that repeats.

Find Duplicate in Array.

Given a read only array of n integers between 1 and n , find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main(){
3     int n;
4     scanf("%d",&n);
5     int arr[n];
6     int hash[n+1];
7     for(int i=0;i<n;i++){
8         hash[i]=0;
9     }
10    for(int i=0;i<n;i++){
11        scanf("%d",&arr[i]);
12    }
13    for(int i=0;i<n;i++){
14        if(hash[arr[i]]==1){
15            printf("%d",arr[i]);
16        }
17        hash[arr[i]]=1;
18    }
19 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



3-Print Intersection of 2 sorted arrays- $O(m*n)$ Time Complexity, $O(1)$ Space Complexity

Started on	Friday, 9 October 2026, 12:48 AM
State	Finished
Completed on	Friday, 9 October 2026, 1:13 AM
Time taken	24 mins 9 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00 | [Flag question](#)

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

- The first line contains T, the number of test cases. Following T lines contain:
 1. Line 1 contains N1, followed by N1 integers of the first array
 2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1	10 57
3 10 17 57	
6	
2 7 10 15 57 246	

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main(){
3     int T,n1,n2;
4     scanf("%d",&T);
5     while(T--){
6         scanf("%d",&n1);
7         int arr[n1];
8         for(int i=0;i<n1;i++){
9             scanf("%d",&arr[i]);
10        }
11        scanf("%d",&n2);
12        int arr2[n2];
13        for(int i=0;i<n2;i++){
14            scanf("%d",&arr2[i]);
15        }
16        for(int i=0;i<n1;i++){
17            for(int j=0;j<n2;j++){
18                if(arr[i]==arr2[j]){
19                    printf("%d ",arr[i]);
20                    break;
21                }
22                if(arr2[j]>arr[i]){
23                    break;
24                }
25            }
26        }
27        printf("\n");
28    }
29 }
```

	Input	Expected	Got	
✓	1	10 57	10 57	✓
	3 10 17 57			
	6			
	2 7 10 15 57 246			

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



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M2

Dashboard My courses

CS23331-DAA-2025-CSE / 4-Print Intersection of 2 sorted arrays- $O(m+n)$ Time Complexity, $O(1)$ Space Complexity

4-Print Intersection of 2 sorted arrays- $O(m+n)$ Time Complexity, $O(1)$ Space Complexity

Started on	Friday, 9 October 2026, 1:13 AM
State	Finished
Completed on	Friday, 9 October 2026, 2:21 AM

Time taken: 1 hour 7 mins

Time taken	1 hour 7 mins
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

The first line contains T, the number of test cases. Following T lines contain:

- Line 1 contains N_1 , followed by N_1 integers of the first array
- Line 2 contains N_2 , followed by N_2 integers of the second array

Output Format

The intersection of the arrays in a single line

The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
1
2 #include <stdio.h>
3
4 int main() {
5     int T;
6     scanf("%d", &T);
```

```

4 ▾ int main() {
5     int T;
6     scanf("%d", &T);
7
8 ▾     while (T--) {
9         int m, n;
10
11         scanf("%d", &m);
12         int a[m];
13
14 ▾         for (int i = 0; i < m; i++) {
15             scanf("%d", &a[i]);
16         }
17
18         scanf("%d", &n);
19         int b[n];
20
21 ▾         for (int i = 0; i < n; i++) {
22             scanf("%d", &b[i]);
23         }
24
25         int i = 0, j = 0;
26
27 ▾         while (i < m && j < n) {
28 ▾             if (a[i] == b[j]) {
29 ▾                 if(a[i]!=a[i-1]){
30                     printf("%d ", a[i]);
31                 }
32

```

```

31         }
32         int value = a[i];
33
34         while (i < m && a[i] == value) {
35             i++;
36 ▾         }
37         while (j < n && b[j] == value) {
38             j++;
39 ▾         }
40         }
41         }
42         else if (a[i] < b[j]) {
43 ▾             i++;
44         }
45         else {
46 ▾             j++;
47         }
48     }
49     printf("\n");
50 }
51 return 0;
52 }
53
54 }
55
56

```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



5-Pair with Difference- $O(n^2)$ Time Complexity, $O(1)$ Space Complexity

Started on	Friday, 9 October 2026, 2:22 AM
State	Finished
Completed on	Friday, 9 October 2026, 2:30 AM
Time taken	8 mins 31 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00 | [Flag question](#)

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[i] - A[j] = k, i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5	1

1 3 5
4

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main(){
3     int n;
4     scanf("%d",&n);
5     int a[n];
6     for(int i=0;i<n;i++){
7         scanf("%d",&a[i]);
8     }
9     int k;
10    scanf("%d",&k);
11    for(int i=0;i<n;i++){
12        for(int j=i+1;j<n;j++){
13            if(a[j]-a[i]==k){
14                printf("1");
15                return 0;
16            }
17        }
18    }
19    printf("0\n");
20 }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



6-Pair with Difference -O(n) Time Complexity,O(1) Space Complexity

Started on	Friday, 9 October 2026, 2:31 AM
State	Finished
Completed on	Friday, 9 October 2026, 2:59 AM
Time taken	28 mins 5 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00 | [Flag question](#)

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result

Input	Result
3	1
1 3 5	
4	

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      int arr[n];
6      for(int i=0;i<n;i++){
7          scanf("%d",&arr[i]);
8      }
9      int k;
10     scanf("%d",&k);
11     int i=0,j=1;
12     while(i<n&&j<n){
13         if(i==j){
14             j++;
15             continue;
16         }
17         int diff=arr[j]-arr[i];
18         if(diff==k){
19             printf("1\n");

```

```

10     scanf("%d",&k);
11     int i=0,j=1;
12     while(i<n&&j<n){
13         if(i==j){
14             j++;
15             continue;
16         }
17         int diff=arr[j]-arr[i];
18         if(diff==k){
19             printf("1\n");
20             return 0;
21         }
22         else if(diff<k){
23             j++;
24         }
25         else{
26             i++;
27         }
28     }
29     printf("0\n");
30 }
31 }

```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.